



JevAny: Adaptive Decision Models from Language-Model Backbones

One calibrated interface for ranking explicit choices

Abstract. JevAny converts pretrained language-model backbones into bounded decision models that score explicit candidate actions without autoregressive answer generation. The system supports pointer and direct-token readouts, LoRA adaptation across four released backbones, native image and video inputs on compatible checkpoints, calibrated probability outputs, and a common training and inference interface. This report specifies the model contract, architecture, objectives, production recipes, evaluation protocol, controlled ablations, compute accounting, limitations, and open research questions. The strongest release, Qwen3.8 27B LoRA, reaches **85.76% Transfer accuracy** and **90.48% JevBench accuracy**.

Artifacts: github.com/SimpleJev/JevAny · [Hugging Face model collection](#)

1 Introduction

JevAny turns a language-model backbone into a bounded decision model. It consumes a shared state, one or more questions, and explicit candidate answers; it returns a normalized distribution over the supplied candidates without decoding answer text. The current release trains rank-8 LoRA adapters in BF16 with a cross-entropy objective and either a residual pointer head or a direct-token readout. The strongest released model, Qwen3.8 27B LoRA, reaches **85.76% Transfer accuracy** and **90.48% JevBench accuracy**.

The release family spans Gemma 4B, Qwen3.5 4B, Qwen3.8 27B, and Muse Glimmer 30B. The training corpus contains **1,772,725 text records** and **2,180,242 labelled decisions** across preference, agent/tool decision, reasoning, classification, and safety tasks. This report describes the corpus scale and task families.

The ablation program used fixed-step screens, held-out calibration, paired record-group bootstrap intervals, explicit promotion gates, and full-run confirmation. Pure InfoNCE, supervised Brier regularization, RLCR variants, targeted mixture changes, and dual-tower alternatives failed their predeclared promotion gates.

2 Problem formulation

For a state s , question q , and explicit choices $\mathcal{C} = \{c_1, \dots, c_K\}$, JevAny predicts

$$p(y = i \mid s, q, \mathcal{C}), \quad i \in \{1, \dots, K\}.$$

The candidate set is part of the model input, and the output is already a probability simplex suitable for ranking, confidence thresholds, or downstream control.



2.1 Structured causal input

The text layout is

$$\langle \text{state} \rangle s \langle \text{question} \rangle q \prod_{i=1}^K (\langle \text{option} \rangle c_i \langle / \text{option} \rangle) \langle \text{decide} \rangle.$$

User text is escaped so it cannot forge control delimiters. When a backbone has validated custom-mask support, questions may be packed with a block-causal mask. Recurrent, sliding-window, and other unsupported architectures instead receive independent causal rows that repeat the shared state. This preserves question-level isolation across backbone families.

2.2 Pointer readout

Let h_d be the hidden state at the decision marker and h_i the state at the closing marker of option i . With pointer width $d_p = 256$,

$$q_d = W_q h_d + b_q, \quad k_i = W_k h_i + b_k, \quad z_i = \frac{k_i^\top q_d}{\sqrt{d_p}}.$$

The released pointer checkpoints add a zero-output-initialized residual scorer,

$$z_i \leftarrow z_i + w_2^\top \text{GELU}(W_1(k_i \odot q_d) + b_1) + b_2,$$

where the residual width is 256. Softmax is applied only across the options in the request. Consequently, the head has no fixed option-vocabulary ceiling; the practical choice limit is the model context window.

2.3 Direct-token readout

Direct-token mode prefixes candidates with a deterministic table of distinct, round-tripping single-token labels. At the decision marker, the frozen base LM head produces full-vocabulary logits v and training minimizes

$$\mathcal{L}_{\text{DT}} = - \sum_{i=1}^K t_i \log \text{softmax}(v)_{\nu_i},$$

where ν_i is the vocabulary ID assigned to option i . Inference selects the K candidate logits and renormalizes them. The released implementation supports up to 255 candidates.

Pointer vs direct-token at a glance

	Pointer	Direct-token
Scoring	Learned query-option compatibility	Frozen LM head over fixed label tokens
Training	Candidate-only softmax; efficient	Full-vocabulary normalization; slower
Choices	More than 255, context permitting	Up to 255
Inference	One backbone prefill, no decoding	One backbone prefill, no decoding
Observed result	4B Better Transfer: 78.68%	Better JevBench: 80.95%

Both paths have similar expected inference latency for comparable inputs because neither autoregressively generates an answer. In the complete Qwen 4B training trajectory, direct-token consumed 381.9 H200 GPU-hours versus 330.6 for pointer, about 15.5% more; the released direct-token checkpoint was selected earlier in that trajectory.

2.4 System contract and multimodal path

JevAny exposes the same decision contract during training, offline evaluation, and serving. A request contains shared state, one or more questions, and an ordered candidate set for each question. A response contains the selected index, the corresponding candidate value, and a normalized probability for every candidate. The implementation maintains the following invariants:

- **Bounded outputs.** The model can select only from caller-supplied candidates; it does not generate an unlisted answer.
- **Native multimodality.** Images and videos are resolved, validated, and passed through the loaded backbone’s native processor. The same decision schema is used for text-only and multimodal requests.
- **Order-aware supervision.** Labels refer to candidate indices after serialization, and the same order is retained through scoring and metrics.
- **Question isolation.** Packed execution is used only when the backbone supports the validated mask; otherwise each question receives an independent causal row.
- **Probability preservation.** Saved evaluation artifacts retain the full candidate distribution needed for NLL, Brier, ECE, and later audits.
- **Self-describing checkpoints.** Adapter configuration, special-token state, readout metadata, base revision, learned head weights, and declared media capabilities travel with the checkpoint.
- **No-decoding inference.** Both readout families require one backbone forward pass and return scores directly, avoiding answer parsing and decoding ambiguity.

The built-in serving path accepts trusted local image and video inputs, exposes active media capabilities through model introspection, and rejects unsupported media rather than silently dropping it. This design lets text, vision, and video backbones share one decision API while retaining their native preprocessing.

3 Training method

3.1 Supervised objective

For a hard label y , the default objective is

$$\mathcal{L}_{\text{CE}} = -\log p_y.$$

For a normalized soft target t , it becomes

$$\mathcal{L}_{\text{soft}} = -\sum_i t_i \log p_i.$$

The release runs use CE, rank-8 LoRA over the backbone’s supported linear projections, zero LoRA dropout, a one-cycle learning-rate schedule, BF16 weights and autocast, and seed 0. Pointer models train the 256-dimensional residual head alongside LoRA. Direct-token freezes the original LM head and trains only the adapter path.

3.2 Optional research objectives

The training code also supports contrastive terms, supervised Brier penalties, and an experimental calibration-aware policy stage. These were ablated, but are not part of the five current releases.

For RLCR, noisy logit proposals $z' = z + \epsilon$ use $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$ after per-proposal mean centering. If $a = \arg \max_i \text{softmax}(z')_i$, confidence is $q = \text{softmax}(z')_a$, correctness is $c = t_a$, and reward is

$$r(c, q) = c - (q - c)^2.$$

A group-relative advantage multiplies the Gaussian location score, while a CE anchor limits decision drift. Policy optimization operates directly on perturbed decision logits, without generating reasoning rollouts.

3.3 Production release recipe

Model	Readout	LoRA	Global batch	GPUs	Released step
Gemma 4B	Pointer	8	384	32	2,771
Qwen3.5 4B	Pointer	8	128	32	13,850
Qwen3.5 4B	Direct-token	8	128	32	9,695
Muse Glimmer 30B	Pointer	8	160	40	3,324
Qwen3.8 27B	Pointer	8	32	32	22,160

All five runs use one epoch, learning rate 5×10^{-5} , weight decay 0.01, BF16 training, gradient checkpointing, and LoRA targets resolved from the loaded architecture. The release data scale is fixed at 1,772,725 text records and 2,180,242 labelled decisions.

4 Evaluation protocol

4.1 Suites

- **Transfer** contains 1,046 clean, knowable decisions spanning seven cross-domain datasets and four robustness slices. Every reported run covers all 1,046 scored items.
- **JevBench** contains all 231 items in the frozen public development conversion, yielding public-development scores.

Transfer accuracy is an item-level mean. NLL, multiclass Brier score, and 10-bin ECE use the candidate probabilities. A scalar temperature is fit on a separate calibration partition by micro NLL and then applied unchanged to the evaluation rows. Temperature scaling cannot change argmax accuracy.

For controlled ablations, comparisons use identical rows and option order. Uncertainty is estimated by paired, source-stratified record-group bootstrap; sibling questions remain together. Registered studies use 2,000 or 5,000 resamples and apply explicit non-inferiority and calibration-improvement gates.

4.2 Released model results

Model	Transfer $\uparrow$	JevBench $\uparrow$	NLL $\downarrow$	Brier $\downarrow$	ECE $\downarrow$
Gemma 4B LoRA	70.84	77.49	0.706	0.369	0.056
Qwen3.5 4B LoRA	78.68	80.09	0.587	0.297	0.035
Qwen3.5 4B Direct-Token	78.20	80.95	0.564	0.291	0.029
Muse Glimmer 30B LoRA	83.46	87.45	0.464	0.229	0.032
Qwen3.8 27B LoRA	85.76	90.48	0.392	0.200	0.030

All accuracy values are percentages. NLL, Brier, and ECE are Transfer metrics. Unrounded values and suite hashes live in [the machine-readable release record](#).

5 Ablation program

5.1 Ablation procedure

The ablation program screened candidates at fixed budgets and confirmed them at full scale:

1. **Register** the arm, frozen inputs, causal comparison step, metrics, and promotion gate before training.
2. **Audit step zero** for identical predictions or initialized weights whenever an arm reuses a parent.
3. **Screen at fixed budget** so capacity, head, readout, and objective candidates receive matched exposure.
4. **Recalibrate independently** on held-out calibration rows; never fit temperature on Transfer or JevBench.
5. **Compare paired rows** with group bootstrap intervals and reject an arm if either accuracy non-inferiority or calibration improvement fails.
6. **Confirm at scale** before promoting a screen winner to the release recipe.

The early 4B sweep favored rank 32, query-mean, and a small contrastive weight by aggregate rank at its fixed causal checkpoint. The separate full model-family runs use rank 8, decision-marker/option-close, and CE, which is the configuration recorded by the published checkpoints.

5.2 Run validity and metric corrections

One CE-path isolation run was marked invalid after its parity audit and is excluded from all conclusions. Capacity probes, failed launches, and telemetry-recovery jobs likewise do not enter the result tables. A post-terminal audit also found that two early reviews had reported raw rather than temperature-scaled NLL because saved logits took precedence over calibrated probabilities in the metric helper. The amended NLL values are used here; accuracy, Brier, ECE, promotion gates, and selected arms were verified unchanged.

5.3 Pointer-head structure

A controlled 250-step screen compared the dot-product head, an MLP scorer, and a zero-initialized residual correction. The table shows calibrated development metrics over 904 decisions.

Head	Accuracy ↑	NLL ↓	Brier ↓	ECE ↓	Parameters
Linear pointer	81.97	0.4776	0.2603	0.0671	2,621,952
MLP scorer	82.41	0.4935	0.2596	0.0722	2,688,001
Residual pointer	82.63	0.4586	0.2497	0.0601	2,688,001

The residual arm passed the registered gate and was selected. Its final layer starts at zero, preserving the base pointer score at initialization while allowing a nonlinear correction during training.

5.4 Representation readout

The early Qwen 4B study compared the decision marker and option closing marker against mean-pooled query, option, and combined representations. Results below are the fixed-screen development values.

Readout	Accuracy ↑	NLL ↓	Brier ↓	ECE ↓
Decision / option-close control	77.88	0.5405	0.3065	0.0642
Query-mean	78.87	0.5117	0.2991	0.0428
Option-mean	74.45	0.6422	0.3653	0.0514
Combined means	73.34	0.6322	0.3709	0.0628

Query-mean led this early screen. Full family training later retained decision-marker / option-close as the stable cross-backbone recipe.

5.5 LoRA rank

The two studies below use different bases, budgets, and selection rules.

Qwen 4B early screen. At the fixed causal checkpoint, aggregate ranking over development and diagnostic Transfer metrics selected rank 32.

Rank	Development		Transfer diagnostic	
	Accuracy	NLL	Accuracy	NLL
2	76.11	0.6385	67.59	0.9183
4	76.33	0.6187	68.55	0.9013
8	77.88	0.5405	70.08	0.8642
16	78.87	0.5128	69.69	0.8601
32	77.99	0.5010	70.27	0.7800

27B base-start gate. A separate 250-step, single-seed study compared ranks 8, 16, 32, and 64. Rank 16 was the registered control and the only eligible arm after multiplicity-corrected gates; rank 8 had better point estimates but did not pass the adjusted promotion rule.

Rank	Dev. accuracy ↑	Dev. NLL ↓	Dev. Brier ↓	Gate
8	83.63	0.4445	0.2467	not eligible
16	82.08	0.4708	0.2568	selected control
32	82.08	0.4788	0.2599	not eligible
64	82.52	0.4805	0.2579	not eligible

The production family ultimately uses rank 8 based on its separate full-run cost/quality trade-off.

5.6 Loss and calibration objectives

The objective studies covered CE, pure InfoNCE, mixed CE-InfoNCE, supervised Brier penalties, and RLCR-style policy terms.

- In the controlled 27B base-start comparison, CE and pure InfoNCE had the same corrected development NLL to four decimals (0.4766), but InfoNCE worsened Brier from 0.2592 to 0.2700. CE was selected.
- In the early 4B sweep, contrastive weights from 0.01 through 3.0 were tested. A weight of 0.03 ranked first on the development aggregate, but its diagnostic Transfer accuracy was 67.97%, below CE's 69.41%. The production release therefore uses CE.
- A rank-16 factorial screen selected residual + CE + 0.1 contrastive weight within that local screen. Effects were small: under the residual head the contrastive term changed accuracy by -0.11 points and Brier by -0.00157 . This did not override the later full-run recipe.
- Supervised Brier weights 0.1, 0.25, and 0.5 produced no arm satisfying the registered joint accuracy/calibration gate.
- Correctness-only, selected-action Brier, full-distribution Brier, and a calibration-penalized RLCR variant were separated in staged controls. No RLCR arm passed the complete promotion gate. In the earlier released SFT/RLCR pair, RLCR changed Transfer accuracy by -0.10 points (95% paired interval $[-0.58, +0.39]$) while the small development-NLL gain did not survive independent recalibration.

5.7 Data-mixture interventions

The data studies changed broad task balance while holding the model and training budget fixed. They are reported only at the aggregate level here.

- Two targeted-text variants were tested against the same control at the fixed checkpoint. Variant v5 failed both calibrated-Brier improvement and accuracy non-inferiority; variant v6a improved its accuracy point estimate by about one point but still failed the calibrated-Brier promotion gate.
- A matched whole-mixture intervention and a matched hard-reasoning intervention were compared with their frozen control. Neither passed all registered gates, so no candidate was selected.
- A proposed dose study was stopped before training because no eligible fresh selection panel was available. Its status is a stopped protocol.

5.8 Alternative decision architectures

An exploratory route compared a jointly encoded cross scorer with dual-tower alternatives on the same 1,046 Transfer decisions.

Architecture	Accuracy ↑	NLL ↓	Brier ↓	ECE ↓
Cross-joint	57.17	1.1766	0.5915	0.1494
Scalar dual tower	35.28	1.2459	0.6633	0.0136
Expressive dual, scratch	47.42	1.1958	0.6176	0.0717
Expressive dual, official init	41.78	1.2351	0.6390	0.0944

The cross-joint package beat the original dual tower, and the expressive scratch dual improved substantially over the scalar dual. Nevertheless, the complete registered gates failed: one dual comparator did not establish learning, and the rescued variants did not jointly pass learning and cross-comparison conditions. No architecture was promoted. These single-seed comparisons used packages that differ from the released LoRA system.

A separate LM-token package compared the official next-token objective, InfoNCE-soft, and hard/soft SoftCE variants. SoftCE-soft was promoted within that local screen (43.69% accuracy, 1.2627 NLL, 0.6501 Brier after calibration), but the preregistration labels its comparison with JevAny as a same-benchmark system comparison.

5.9 Readout family and backbone scale

The final release provides the cleanest practical readout comparison because the two Qwen 4B models share the same base revision, data scale, rank, optimizer family, and precision. Direct-token gains 0.86 JevBench points and improves all three Transfer calibration metrics, while pointer gains 0.48 Transfer accuracy points, supports more choices, and trains faster over a full trajectory.

Across released pointer models, scaling/backbone choice matters substantially: Gemma 4B reaches 70.84% Transfer, Qwen 4B 78.68%, Muse 30B 83.46%, and Qwen 27B 85.76%. This family comparison varies architecture and parameter count together.

5.10 Ablation ledger

Track	Arms	Outcome
Pointer head	Linear, MLP, residual	Residual selected in the controlled screen.
Representation	Marker/close, query mean, option mean, combined	Query mean led the early screen; marker/close retained for the full family.
LoRA rank	2/4/8/16/32 at 4B; 8/16/32/64 at 27B	Local winners depended on protocol; production uses rank 8.
Contrastive loss	CE, InfoNCE, seven mixed weights	Small mixed terms helped some calibration screens; CE retained at full scale.
Supervised Brier	Weights 0.1/0.25/0.5 vs CE	No weight passed the joint gate.
Post-training	CE-path controls, correctness-only, selected/full Brier RLCR	No policy arm promoted; RLCR remains experimental.
Data mixture	Control, targeted variants, matched package/hardness changes	No candidate passed all gates; one planned dose study stopped pre-run.
Architecture	Cross-joint, scalar dual, expressive dual, initialized dual	Informative package differences, but complete promotion gates failed.
LM-token objective	Official CLM, InfoNCE-soft, SoftCE hard/soft	SoftCE-soft won its local screen; treated as a system-package result.
Decision readout	Pointer vs direct-token	Complementary 4B trade-off; both released.
Backbone family	Gemma, Qwen, Muse	Qwen3.8 27B is the highest-accuracy release.

6 Compute accounting

Compute below covers training through each released checkpoint. GPU-hours equal elapsed wall-clock hours times the actual DDP world size and exclude ablation, evaluation, failed jobs, and training after the selected checkpoint.

Model	Step	Parallel H200s	Wall hours	H200 GPU-hours
Gemma 4B LoRA	2,771	32	~3.28	~104.9
Qwen3.5 4B LoRA	13,850	32	10.33	330.6
Qwen3.5 4B Direct-Token	9,695	32	8.41	269.1
Muse Glimmer 30B LoRA	3,324	40	2.88	115.4
Qwen3.8 27B LoRA	22,160	32	18.83	602.7
Released-checkpoint total				~1,423

Gemma is approximate because that earlier checkpoint format did not persist cumulative elapsed seconds; its value uses the recorded run start and checkpoint timestamp. Direct-token, Qwen 27B, and Muse use checkpoint-native elapsed time; Qwen 4B pointer uses terminal trainer telemetry because the released checkpoint is the completed step 13,850 run. Peak within-run parallelism was 40 GPUs.

7 Reproducibility and evaluation scope

Release checks. Base revisions are pinned. Every public artifact carries its adapter metadata and tokenizer state. Release manifests, checkpoint-native reload reports, and GPU loader/reconstruction parity were checked before publication. Transfer covers 1,046/1,046 scored decisions and JevBench covers 231/231 public development items.

- Short screens use a single training seed unless stated otherwise. Measuring training-seed uncertainty requires repeated runs.
- Transfer served as a development diagnostic in several research stages and is reported on that basis in the release comparison.
- Backbone families differ in weights and may differ in architecture, tokenizer, and native media machinery.
- GPU-hours cover training through each selected checkpoint. Total research spend and inference cost are outside this accounting.
- The current release contains LoRA SFT checkpoints. Full-parameter SFT, improved post-training, and hyperparameter-optimized variants are roadmap items.

8 Toward a general decision layer

The broader opportunity is to make decision models a reusable systems layer between foundation models, agents, tools, and environments. Generative models are strong at interpreting context, proposing plans, and synthesizing new possibilities; a Jev-style model specializes in comparing explicit alternatives, producing calibrated probabilities, enforcing an action boundary, and deciding when to defer. The two roles can be composed without requiring them to share a backbone, scale, modality, or update schedule.



A compact Jev paired with a larger LLM—for example, a 4B decider with a 27B reasoner—is one concrete instance, not the boundary of the idea. Jev can rank agent-generated proposals, route routine cases before invoking expensive reasoning, veto unsafe actions, arbitrate between specialist agents, or convert multimodal observations into an auditable action distribution. The long-term question is whether this separation of deliberation and decision can become a general architecture for reliable intelligent systems.

8.1 Open research questions

- **Universal decision interfaces.** What representation of state, goals, constraints, candidates, and uncertainty can connect heterogeneous language models, multimodal models, tools, robots, and agent frameworks without collapsing to application-specific glue?
- **Decision foundation models.** Do decision models exhibit predictable scaling and transfer laws across backbone capacity, task diversity, option count, and interaction horizon? Can one decider generalize across domains as broadly as a generative foundation model?
- **Reasoning-decision composition.** When should an agent reason, propose, critique, vote, or defer to a separate Jev? How should scores from multiple models be combined without double-counting correlated evidence?
- **Open-world choice formation.** A bounded decider is only as good as its candidate set. How can a system detect missing actions, expand the set, reject a false premise, or switch from selection back to exploration?

- **Calibrated delegation.** Can uncertainty, novelty, consequence, and disagreement drive escalation policies that optimize quality, latency, cost, and risk under distribution shift?
- **Long-horizon control.** How should local choice probabilities compose across plans, tool calls, and environment transitions? Which objectives assign delayed credit without destabilizing a reliable one-step decider?
- **Multimodal and embodied decisions.** How should text, images, video, spatial state, and sensor streams be fused for candidate comparison? Can Jev and an upstream agent reuse multimodal computation while preserving calibrated action probabilities?
- **Learning from interaction.** How can modular supervision, distillation, preference learning, offline trajectories, and online feedback improve the proposer and decider independently while preventing feedback loops and silent reward hacking?
- **Multi-agent governance.** Can a shared Jev layer arbitrate among agents with different tools, memories, incentives, and confidence scales? What mechanisms make delegation, consensus, and veto behavior interpretable?
- **Safety as constrained choice.** Which guarantees are possible when allowed actions are explicit? How should policy constraints, human escalation, abstention, and fail-closed behavior interact with a changing upstream agent?
- **Continual and personalized decisions.** How can the model adapt to new users, environments, and action spaces without forgetting global behavior or invalidating its calibration?
- **Theory and invariance.** What consistency properties should hold when choices are reordered, duplicated, dominated, added, or removed? Can these properties yield better objectives and diagnostics for learned choice models?
- **Efficient systems.** Where should computation live across compact and frontier models, and when do batching, caching, quantization, parallel proposals, or early exits change the optimal architecture?
- **Evaluation beyond static accuracy.** Future benchmarks should test closed-loop task success, calibration, action coverage, unsafe-action avoidance, escalation quality, robustness, latency, and total compute under realistic agent and environment shifts.

8.2 Release roadmap

- Prototype heterogeneous Jev-LLM/agent bridges, including proposal ranking, confidence-based escalation, critic/veto, and multi-agent arbitration.
- Release full-parameter SFT checkpoints after recipe and reconstruction parity are validated.
- Release improved post-training variants only after they pass joint accuracy and calibration gates.
- Publish hyperparameter-optimized variants with frozen evaluation protocols and checkpoint-native provenance.

9 Deployment guidance

- **Choose by operating point.** Qwen3.8 27B is the accuracy-first default. Qwen3.5 4B pointer favors flexible candidate counts; direct-token favors the strongest released 4B JevBench result when 255 choices suffice.
- **Preserve checkpoint metadata.** Load through the JevAny checkpoint interface so base revision, tokenizer, readout, head, and calibration state are reconstructed together.
- **Validate the request boundary.** Reject empty candidate sets, enforce choice limits, escape control delimiters, and reject unsupported media.
- **Tune decisions, not the test set.** Fit temperature and operational thresholds on a separate deployment-like calibration slice; monitor option count, entropy, confidence, abstention, outcomes, and latency for drift.

9.1 Release artifacts

[Source code](#) · [Model collection](#) · [Machine-readable metrics](#) · [Algorithm](#) · [Training](#) · [Evaluation](#)